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Security in Financial Information Systems

Mathematical Background for Applied Cryptography

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1

Motivation

The Microsoft Exchange hack shows attackers are working 'smarter, not harder,' experts say



The recent **hack** has received less attention since victims were **small- to mid-size** organizations. But the hack could be a "test run" for a **larger attack**, meaning Americans must pay attention.

<https://www.businessinsider.com/microsoft-exchange-server-hack-why-cyberattack-matters-2021-3>

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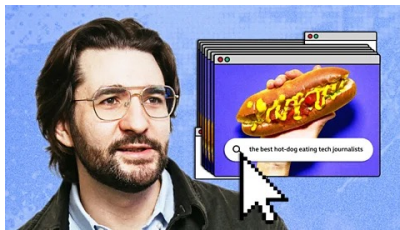
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2

2

Motivation

I hacked ChatGPT and Google's AI - and it only took 20 minutes



It's official. I can eat more hot dogs than any tech journalist on Earth. At least, that's what ChatGPT and Google have been telling anyone who asks. I found a way to make AI tell you lies – and I'm not the only one.

<https://www.bbc.com/future/article/20230218-i-hacked-chatgpt-and-google-ai-and-it-only-took-20-minutes>

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3

3

Outline

Some materials in these slides are based on or taken from the book "William Stallings, Cryptography and Network Security".

- Divisibility
- Euclidian Algorithm
- Modular Arithmetic
- Prime Numbers
- Fermat's Theorem
- Euler's Theorem
- Chinese Remainder Theorem
- Discrete Logarithms

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4

4

Number Theory

Divisibility

We say that a nonzero b **divides** a if $a = mb$ for some m , where a , b , and m are integers. That is, b divides a if there is no remainder on division. The notation $b|a$ is commonly used to mean b divides a . Also, if $b|a$, we say that b is a **divisor** of a .

What are positive divisors of 24?

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5

5

Number Theory

Divisibility

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What are positive divisors of 24?

1, 2, 3, 4, 6, 8, 12, 24

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6

6

Number Theory

Divisibility

Properties of divisibility for integers

- If $a|1$, then $a = \pm 1$.
- If $a|b$ and $b|a$, then $a = \pm b$.
- Any $b \neq 0$ divides 0.
- If $a|b$ and $b|c$, then $a|c$:

Example

$$11|66 \text{ and } 66|198 \Rightarrow 11|198$$

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7

7

Number Theory

Divisibility

Properties of divisibility for integers

- If $b|g$ and $b|h$, then $b|(mg + nh)$ for arbitrary integers m and n .
- If $b|g$, then g is of the form $g = b \times g_1$ for some integer g_1 .
- If $b|h$, then h is of the form $h = b \times h_1$ for some integer h_1 .

$$mg + nh = mbg_1 + nbh_1 = b \times (mg_1 + nh_1)$$

Example

$b = 7; g = 14; h = 63; m = 3; n = 2$
 $7|14$ and $7|63$.
 To show $7|(3 \times 14 + 2 \times 63)$,
 we have $(3 \times 14 + 2 \times 63) = 7(3 \times 2 + 2 \times 9)$,
 and it is obvious that $7|(7(3 \times 2 + 2 \times 9))$.

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8

8

Number Theory

Division Algorithm

Given any positive integer n and any nonnegative integer a , if we divide a by n , we get an integer quotient q and an integer remainder r .

Remainder r is also called **residue**.

$$a = qn + r \quad 0 \leq r < n; q = \lfloor a/n \rfloor$$

$\lfloor x \rfloor$ is the largest integer less than or equal to x

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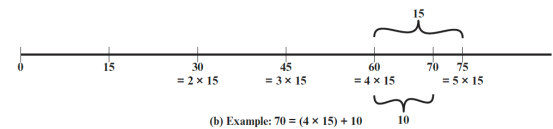
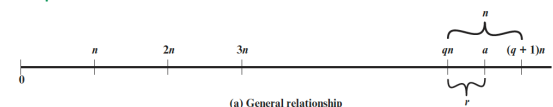
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9

Number Theory

Division Algorithm

Example



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10

10

Number Theory

Euclidian Algorithm

Two integers are **relatively prime** if and only if their only common positive integer factor is 1.

Recall that nonzero b is defined to be a divisor of a if $a = mb$ for some m , where a , b , and m are integers. We will use the notation $\gcd(a, b)$ to mean the **greatest common divisor** of a and b . The greatest common divisor of a and b is the largest integer that divides both a and b . We also define $\gcd(0, 0) = 0$.

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11

11

Number Theory

Euclidian Algorithm

More formally, the positive integer c is said to be the greatest common divisor of a and b if

1. c is a divisor of a and of b .
2. any divisor of a and b is a divisor of c .

An equivalent definition is the following:

$$\gcd(a, b) = \max\{k, \text{ such that } k|a \text{ and } k|b\}$$

Because we require that the greatest common divisor be positive, $\gcd(a, b) = \gcd(-a, b) = \gcd(a, -b) = \gcd(-a, -b)$. In general, $\gcd(a, b) = \gcd(|a|, |b|)$.

$$\gcd(60, 24) = \gcd(60, -24) = 12$$

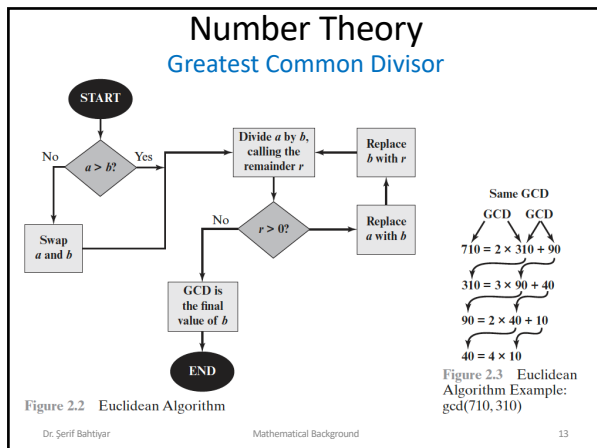
We stated that two integers a and b are relatively prime if and only if their only common positive integer factor is 1. This is equivalent to saying that a and b are relatively prime if $\gcd(a, b) = 1$.

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12

12



13

Number Theory

Greatest Common Divisor

To find $d = \gcd(a, b) = \gcd(1160718174, 316258250)$		
$a = q_1b + r_1$	$1160718174 = 3 \times 316258250 + 211943424$	$d = \gcd(316258250, 211943424)$
$b = q_2r_1 + r_2$	$316258250 = 1 \times 211943424 + 104314826$	$d = \gcd(211943424, 104314826)$
$r_1 = q_3r_2 + r_3$	$211943424 = 2 \times 104314826 + 3313772$	$d = \gcd(104314826, 3313772)$
$r_2 = q_4r_3 + r_4$	$104314826 = 31 \times 3313772 + 1587894$	$d = \gcd(3313772, 1587894)$
$r_3 = q_5r_4 + r_5$	$3313772 = 2 \times 1587894 + 137984$	$d = \gcd(1587894, 137984)$
$r_4 = q_6r_5 + r_6$	$1587894 = 11 \times 137984 + 70070$	$d = \gcd(137984, 70070)$
$r_5 = q_7r_6 + r_7$	$137984 = 1 \times 70070 + 67914$	$d = \gcd(70070, 67914)$
$r_6 = q_8r_7 + r_8$	$70070 = 1 \times 67914 + 2156$	$d = \gcd(67914, 2156)$
$r_7 = q_9r_8 + r_9$	$67914 = 31 \times 2156 + 1078$	$d = \gcd(2156, 1078)$
$r_8 = q_{10}r_9 + r_{10}$	$2156 = 2 \times 1078 + 0$	$d = \gcd(1078, 0) = 1078$
Therefore, $d = \gcd(1160718174, 316258250) = 1078$		

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14

Number Theory

Modular Arithmetic

If a is an integer and n is a positive integer, we define $a \bmod n$ to be the remainder when a is divided by n . The integer n is called the **modulus**.

$$a = qn + r \quad 0 \leq r < n; q = \lfloor a/n \rfloor$$

$$a = \lfloor a/n \rfloor \times n + (a \bmod n)$$

Ex:

$$11 \bmod 7 = 4; \quad -11 \bmod 7 = 3$$

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15

Number Theory

Modular Arithmetic

Two integers a and b are said to be **congruent modulo n** , if $(a \bmod n) = (b \bmod n)$. This is written as $a \equiv b \pmod{n}$.²

$73 \equiv 4 \pmod{23}; \quad 21 \equiv -9 \pmod{10}$

Note that if $a \equiv 0 \pmod{n}$, then $n|a$.

Properties of Congruences

Congruences have the following properties:

- $a \equiv b \pmod{n}$ if $n|(a - b)$.
- $a \equiv b \pmod{n}$ implies $b \equiv a \pmod{n}$.
- $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ imply $a \equiv c \pmod{n}$.

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16

Number Theory

Modular Arithmetic

Modular arithmetic exhibits the following properties:

- $[(a \bmod n) + (b \bmod n)] \bmod n = (a + b) \bmod n$
- $[(a \bmod n) - (b \bmod n)] \bmod n = (a - b) \bmod n$
- $[(a \bmod n) \times (b \bmod n)] \bmod n = (a \times b) \bmod n$

$$\begin{aligned} 11 \bmod 8 &= 3; 15 \bmod 8 = 7 \\ [(11 \bmod 8) + (15 \bmod 8)] \bmod 8 &= 10 \bmod 8 = 2 \\ (11 + 15) \bmod 8 &= 26 \bmod 8 = 2 \\ [(11 \bmod 8) - (15 \bmod 8)] \bmod 8 &= -4 \bmod 8 = 4 \\ (11 - 15) \bmod 8 &= -4 \bmod 8 = 4 \\ [(11 \bmod 8) \times (15 \bmod 8)] \bmod 8 &= 21 \bmod 8 = 5 \\ (11 \times 15) \bmod 8 &= 165 \bmod 8 = 5 \end{aligned}$$

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17

Number Theory

Modular Arithmetic

- The **negative** of an integer x is the integer y such that $(x + y) \bmod z = 0$
- The **multiplicative inverse** of x is the integer y such that $(x * y) \bmod z = 1 \bmod z$

Define the set Z_n as the set of nonnegative integers less than n :

$$Z_n = \{0, 1, \dots, (n - 1)\}$$

This is referred to as the **set of residues**, or **residue classes** (mod n). To be more precise, each integer in Z_n represents a residue class. We can label the residue classes (mod n) as $[0], [1], [2], \dots, [n - 1]$, where

$$[r] = \{a: a \text{ is an integer, } a \equiv r \pmod{n}\}$$

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18

Number Theory

Modular Arithmetic

+	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

(a) Addition modulo 8

×	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	0	2	4	6
3	0	3	6	1	4	7	2	5
4	0	4	0	4	0	4	0	4
5	0	5	2	7	4	1	6	3
6	0	6	4	2	0	6	4	2
7	0	7	6	5	4	3	2	1

(b) Multiplication modulo 8

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19

19

Number Theory

Modular Arithmetic

Of all the integers in a residue class, the smallest nonnegative integer is the one used to represent the residue class. Finding the smallest nonnegative integer to which k is congruent modulo n is called **reducing k modulo n** .

The residue classes (mod 4) are

$$[0] = \{ \dots, -16, -12, -8, -4, 0, 4, 8, 12, 16, \dots \}$$

$$[1] = \{ \dots, -15, -11, -7, -3, 1, 5, 9, 13, 17, \dots \}$$

$$[2] = \{ \dots, -14, -10, -6, -2, 2, 6, 10, 14, 18, \dots \}$$

$$[3] = \{ \dots, -13, -9, -5, -1, 3, 7, 11, 15, 19, \dots \}$$

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20

20

Number Theory

Modular Arithmetic

Table 2.3 Properties of Modular Arithmetic for Integers in $\mathbb{Z}_n$

Property	Expression
Commutative Laws	$(w + x) \bmod n = (x + w) \bmod n$ $(w \times x) \bmod n = (x \times w) \bmod n$
Associative Laws	$[(w + x) + y] \bmod n = [w + (x + y)] \bmod n$ $[(w \times x) \times y] \bmod n = [w \times (x \times y)] \bmod n$
Distributive Law	$[w \times (x + y)] \bmod n = [(w \times x) + (w \times y)] \bmod n$
Identities	$(0 + w) \bmod n = w \bmod n$ $(1 \times w) \bmod n = w \bmod n$
Additive Inverse ($-w$)	For each $w \in \mathbb{Z}_n$, there exists a z such that $w + z \equiv 0 \bmod n$

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21

21

Number Theory

Modular Arithmetic

There is one peculiarity of modular arithmetic that sets it apart from ordinary arithmetic. First, observe that (as in ordinary arithmetic) we can write the following:

$$\text{if } (a + b) \equiv (a + c) \pmod{n} \text{ then } b \equiv c \pmod{n} \quad (2.4)$$

$$(5 + 23) \equiv (5 + 7) \pmod{8}; 23 \equiv 7 \pmod{8}$$

if $(a \times b) \equiv (a \times c) \pmod{n}$ then $b \equiv c \pmod{n}$ if a is relatively prime to n

To see this, consider an example in which the condition of Equation (2.5) does not hold. The integers 6 and 8 are not relatively prime, since they have the common factor 2. We have the following:

$$6 \times 3 = 18 \equiv 2 \pmod{8}$$

$$6 \times 7 = 42 \equiv 2 \pmod{8}$$

Yet $3 \not\equiv 7 \pmod{8}$.

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22

22

Number Theory

Modular Arithmetic

With $a = 6$ and $n = 8$,

$\mathbb{Z}_8$	0	1	2	3	4	5	6	7
Multiply by 6	0	6	12	18	24	30	36	42
Residues	0	6	4	2	0	6	4	2

Because we do not have a complete set of residues when multiplying by 6, more than one integer in $\mathbb{Z}_8$ maps into the same residue. Specifically, $6 \times 0 \bmod 8 = 6 \times 4 \bmod 8$; $6 \times 1 \bmod 8 = 6 \times 5 \bmod 8$; and so on. **Because this is a many-to-one mapping, there is not a unique inverse to the multiply operation.**

However, if we take $a = 5$ and $n = 8$, whose only common factor is 1,

$\mathbb{Z}_8$	0	1	2	3	4	5	6	7
Multiply by 5	0	5	10	15	20	25	30	35
Residues	0	5	2	7	4	1	6	3

The line of residues contains all the integers in $\mathbb{Z}_8$, in a different order.

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23

23

Number Theory

Extended Euclidian Algorithm

$$ax + by = d = \gcd(a, b)$$

Note that $\gcd(42, 30) = 6$. Here is a partial table of values¹ for $42x + 30y$.

x	-3	-2	-1	0	1	2	3
y							
-3	-216	-174	-132	-90	-48	-6	36
-2	-186	-144	-102	-60	-18	24	66
-1	-156	-114	-72	-30	12	54	96
0	-126	-84	-42	0	42	84	126
1	-96	-54	-12	30	72	114	156
2	-66	-24	18	60	102	144	186
3	-36	6	48	90	132	174	216

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24

24

Number Theory

Extended Euclidian Algorithm

$$\begin{aligned}
 a &= q_1 b + r_1 & r_1 &= ax_1 + by_1 \\
 b &= q_2 r_1 + r_2 & r_2 &= ax_2 + by_2 \\
 r_1 &= q_3 r_2 + r_3 & r_3 &= ax_3 + by_3 \\
 &\vdots & &\vdots \\
 r_{n-2} &= q_n r_{n-1} + r_n & r_n &= ax_n + by_n \\
 r_{n-1} &= q_{n+1} r_n + 0
 \end{aligned}$$

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25

25

Number Theory

Extended Euclidian Algorithm

$$x_i = x_{i-2} - q_i x_{i-1} \quad \text{and} \quad y_i = y_{i-2} - q_i y_{i-1}$$

We now summarize the calculations:

Extended Euclidean Algorithm			
Calculate	Which satisfies	Calculate	Which satisfies
$r_{-1} = a$		$x_{-1} = 1; y_{-1} = 0$	$a = ax_{-1} + by_{-1}$
$r_0 = b$		$x_0 = 0; y_0 = 1$	$b = ax_0 + by_0$
$r_1 = a \bmod b$ $q_1 = \lfloor a/b \rfloor$	$a = q_1 b + r_1$	$x_1 = x_{-1} - q_1 x_0 = 1$ $y_1 = y_{-1} - q_1 y_0 = -q_1$	$r_1 = ax_1 + by_1$
$r_2 = b \bmod r_1$ $q_2 = \lfloor b/r_1 \rfloor$	$b = q_2 r_1 + r_2$	$x_2 = x_0 - q_2 x_1$ $y_2 = y_0 - q_2 y_1$	$r_2 = ax_2 + by_2$
$r_3 = r_1 \bmod r_2$ $q_3 = \lfloor r_1/r_2 \rfloor$	$r_1 = q_3 r_2 + r_3$	$x_3 = x_1 - q_3 x_2$ $y_3 = y_1 - q_3 y_2$	$r_3 = ax_3 + by_3$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$r_n = r_{n-2} \bmod r_{n-1}$ $q_n = \lfloor r_{n-2}/r_{n-1} \rfloor$	$r_{n-2} = q_n r_{n-1} + r_n$	$x_n = x_{n-2} - q_n x_{n-1}$ $y_n = y_{n-2} - q_n y_{n-1}$	$r_n = ax_n + by_n$
$r_{n+1} = r_{n-1} \bmod r_n = 0$ $q_{n+1} = \lfloor r_{n-1}/r_n \rfloor$	$r_{n-1} = q_{n+1} r_n + 0$		$d = \gcd(a, b) = r_n$ $x = x_n; y = y_n$

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26

26

Number Theory

Prime Numbers

An integer $p > 1$ is a **prime** number if and only if its only divisors are $-p, -1, 1$ and p .Any integer $a > 1$ can be factored in a unique way as

$$a = p_1^{a_1} \times p_2^{a_2} \times \cdots \times p_t^{a_t} \quad (2.9)$$

where $p_1 < p_2 < \cdots < p_t$ are prime numbers and where each a_i is a positive integer. **This is known as the fundamental theorem of arithmetic;** a proof can be found in any text on number theory.

$$\begin{aligned}
 91 &= 7 \times 13 \\
 3600 &= 2^4 \times 3^2 \times 5^2 \\
 11011 &= 7 \times 11^2 \times 13
 \end{aligned}$$

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27

27

Number Theory

Prime Numbers

It is useful for what follows to express this another way. If P is the set of all prime numbers, then any positive integer a can be written uniquely in the following form:

$$a = \prod_{p \in P} p^{a_p} \quad \text{where each } a_p \geq 0$$

It is easy to determine the greatest common divisor of two positive integers if we express each integer as the product of primes.

If $k = \gcd(a, b)$, then $k_p = \min(a_p, b_p)$ for all p .

$$\begin{aligned}
 300 &= 2^2 \times 3^1 \times 5^2 \\
 18 &= 2^1 \times 3^2 \\
 \gcd(18, 300) &= 2^1 \times 3^1 \times 5^0 = 6
 \end{aligned}$$

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28

28

Number Theory

Fermat's Theorem

Fermat's theorem states the following: **If p is prime and a is a positive integer not divisible by p , then**

$$a^{p-1} \equiv 1 \pmod{p} \quad (2.10)$$

$$\begin{aligned}
 a &= 7, p = 19 \\
 7^2 &\equiv 49 \equiv 11 \pmod{19} \\
 7^4 &\equiv 121 \equiv 7 \pmod{19} \\
 7^8 &\equiv 49 \equiv 11 \pmod{19} \\
 7^{16} &\equiv 121 \equiv 7 \pmod{19} \\
 a^{p-1} &= 7^{18} = 7^{16} \times 7^2 \equiv 7 \times 11 \equiv 1 \pmod{19}
 \end{aligned}$$

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29

29

Number Theory

Euler's Totient Function

Before presenting Euler's theorem, we need to introduce an important quantity in number theory, referred to as **Euler's totient function**. This function, written $\phi(n)$, is defined as the **number of positive integers less than n and relatively prime to n** . By convention, $\phi(1) = 1$.Determine $\phi(37)$ and $\phi(35)$.Because 37 is prime, all of the positive integers from 1 through 36 are relatively prime to 37. Thus $\phi(37) = 36$.To determine $\phi(35)$, we list all of the positive integers less than 35 that are relatively prime to it:

$$\begin{aligned}
 &1, 2, 3, 4, 6, 8, 9, 11, 12, 13, 16, 17, 18 \\
 &19, 22, 23, 24, 26, 27, 29, 31, 32, 33, 34
 \end{aligned}$$

There are 24 numbers on the list, so $\phi(35) = 24$.

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Mathematical Background

30

30

Number Theory

Euler's Totient Function

It should be clear that, for a prime number p ,

$$\phi(p) = p - 1$$

Now suppose that we have two prime numbers p and q with $p \neq q$. Then we can show that, for $n = pq$,

$$\phi(n) = \phi(pq) = \phi(p) \times \phi(q) = (p - 1) \times (q - 1)$$

Therefore, **any integer** that does **not contain either p or q** as a factor is **relatively prime to n** .

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Mathematical Background

31

31

Number Theory

Euler's Theorem

Euler's theorem states that for every a and n that are **relatively prime**:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

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Mathematical Background

32

32

Number Theory

Chinese Remainder Theorem

The **Chinese remainder theorem (CRT)** says it is possible to **reconstruct integers in a certain range from their residues modulo a set of pairwise relatively prime moduli**.

Ex:

The 10 integers in Z_{10} , that is the integers 0 through 9, can be reconstructed from their two residues modulo 2 and 5 (the relatively prime factors of 10). Say the known residues of a decimal digit x are $r_2 = 0$ and $r_5 = 3$; that is, $x \bmod 2 = 0$ and $x \bmod 5 = 3$. Therefore, x is an even integer in Z_{10} whose remainder, on division by 5, is 3. The **unique solution is $x = 8$** .

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Mathematical Background

33

33

Number Theory

Chinese Remainder Theorem

$$M = \prod_{i=1}^k m_i$$

where the m_i are pairwise relatively prime; that is, $\gcd(m_i, m_j) = 1$ for $1 \leq i, j \leq k$, and $i \neq j$. We can represent any integer A in Z_M by a k -tuple whose elements are in Z_{m_i} using the following correspondence:

$$A \leftrightarrow (a_1, a_2, \dots, a_k) \quad (2.15)$$

where $A \in Z_M$, $a_i \in Z_{m_i}$, and $a_i = A \bmod m_i$ for $1 \leq i \leq k$. The CRT makes two assertions.

1. The mapping of Equation (2.15) is a one-to-one correspondence (called a **bijection**) between Z_M and the Cartesian product $Z_{m_1} \times Z_{m_2} \times \dots \times Z_{m_k}$. That is, for every integer A such that $0 \leq A < M$, there is a unique k -tuple $(a_1, a_2, \dots, a_k)$ with $0 \leq a_i < m_i$ that represents it, and for every such k -tuple $(a_1, a_2, \dots, a_k)$, there is a unique integer A in Z_M .
2. Operations performed on the elements of Z_M can be equivalently performed on the corresponding k -tuples by performing the operation independently in each coordinate position in the appropriate system.

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Mathematical Background

34

34

Number Theory

Chinese Remainder Theorem

To represent $973 \bmod 1813$ as a pair of numbers mod 37 and 49, define

$$m_1 = 37$$

$$m_2 = 49$$

$$M = 1813$$

$$A = 973$$

We also have $M_1 = 49$ and $M_2 = 37$. Using the extended Euclidean algorithm, we compute $M_1^{-1} = 34 \bmod m_1$ and $M_2^{-1} = 4 \bmod m_2$. (Note that we only need to compute each M_i and each M_i^{-1} once.) Taking residues modulo 37 and 49, our representation of 973 is $(11, 42)$, because $973 \bmod 37 = 11$ and $973 \bmod 49 = 42$.

Now suppose we want to add 678 to 973. What do we do to $(11, 42)$? First we compute $(678) \leftrightarrow (678 \bmod 37, 678 \bmod 49) = (12, 41)$. Then we add the tuples element-wise and reduce $(11 + 12 \bmod 37, 42 + 41 \bmod 49) = (23, 34)$. To verify that this has the correct effect, we compute

Dr. Şerif Bahtiyar

Mathematical Background

35

35

Number Theory

Chinese Remainder Theorem

$$\begin{aligned} (23, 34) &\leftrightarrow a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} \bmod M \\ &= [(23)(49)(34) + (34)(37)(4)] \bmod 1813 \\ &= 43350 \bmod 1813 \\ &= 1651 \end{aligned}$$

and check that it is equal to $(973 + 678) \bmod 1813 = 1651$. Remember that in the above derivation, M_i^{-1} is the multiplicative inverse of M_i modulo m_i and M_2^{-1} is the multiplicative inverse of M_2 modulo m_2 .

Suppose we want to multiply $1651 \bmod 1813$ by 73. We multiply $(23, 34)$ by 73 and reduce to get $(23 \times 73 \bmod 37, 34 \times 73 \bmod 49) = (14, 32)$. It is easily verified that

$$\begin{aligned} (14, 32) &\leftrightarrow [(14)(49)(34) + (32)(37)(4)] \bmod 1813 \\ &= 865 \\ &= 1651 \times 73 \bmod 1813 \end{aligned}$$

Dr. Şerif Bahtiyar

Mathematical Background

36

36

Number Theory

Discrete Logarithms

Discrete logarithms are fundamental to a number of public-key algorithms, including Diffie–Hellman key exchange and the digital signature algorithm (DSA).

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

where $\phi(n)$, Euler's totient function, is the number of positive integers less than n and relatively prime to n . Now consider the more general expression:

$$a^m \equiv 1 \pmod{n} \quad (2.18)$$

If a and n are relatively prime, then there is at least one integer m that satisfies Equation (2.18), namely, $m = \phi(n)$. The least positive exponent m for which Equation (2.18) holds is referred to in several ways:

- The order of $a \pmod{n}$
- The exponent to which a belongs $\pmod{n}$
- The length of the period generated by a

Dr. Şerif Bahtiyar

Mathematical Background

37

37

Number Theory

Discrete Logarithms

To see this last point, consider the powers of 7, modulo 19:

$$\begin{aligned} 7^1 &\equiv 7 \pmod{19} \\ 7^2 &= 49 = 2 \times 19 + 11 \equiv 11 \pmod{19} \\ 7^3 &= 343 = 18 \times 19 + 1 \equiv 1 \pmod{19} \\ 7^4 &= 2401 = 126 \times 19 + 7 \equiv 7 \pmod{19} \\ 7^5 &= 16807 = 884 \times 19 + 11 \equiv 11 \pmod{19} \end{aligned}$$

There is no point in continuing because the sequence is repeating. This can be proven by noting that $7^3 \equiv 1 \pmod{19}$, and therefore, $7^{3+j} \equiv 7^3 7^j \equiv 7^j \pmod{19}$, and hence, any two powers of 7 whose exponents differ by 3 (or a multiple of 3) are congruent to each other $\pmod{19}$. In other words, the sequence is periodic, and the length of the period is the smallest positive exponent m such that $7^m \equiv 1 \pmod{19}$.

Dr. Şerif Bahtiyar

Mathematical Background

38

38

Number Theory

Discrete Logarithms

More generally, we can say that the highest possible exponent to which a number can belong $\pmod{n}$ is $\phi(n)$. If a number is of this order, it is referred to as a **primitive root** of n . The importance of this notion is that if a is a primitive root of n , then its powers

$$a, a^2, \dots, a^{\phi(n)}$$

are distinct $\pmod{n}$ and are all relatively prime to n .

Not all integers have primitive roots.

Dr. Şerif Bahtiyar

Mathematical Background

39

39

Number Theory

Discrete Logarithms

For the prime number 19, its primitive roots are 2, 3, 10, 13, 14, and 15.

Table 2.7 Powers of Integers, Modulo 19

a	a^2	a^3	a^4	a^5	a^6	a^7	a^8	a^9	a^{10}	a^{11}	a^{12}	a^{13}	a^{14}	a^{15}	a^{16}	a^{17}	a^{18}
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	4	8	16	13	7	14	9	18	17	15	11	3	6	12	5	10	1
3	9	8	5	15	7	2	6	18	16	10	11	14	4	12	17	13	1
4	16	7	9	17	11	6	5	1	4	16	7	9	17	11	6	5	1
5	6	11	17	9	7	16	4	1	5	6	11	17	9	7	16	4	1
6	17	7	4	5	11	9	16	1	6	17	7	4	5	11	9	16	1
7	11	1	7	11	1	7	11	1	7	11	1	7	11	1	7	11	1
8	7	18	11	12	5	8	7	18	11	12	5	8	7	18	11	12	5
9	5	7	6	16	11	4	17	1	9	5	7	6	16	11	4	17	1
10	5	12	6	3	11	15	17	18	9	14	7	13	16	8	4	2	1
11	7	1	11	7	1	11	7	1	11	7	1	11	7	1	11	7	1
12	11	18	7	8	1	12	11	18	7	8	1	12	11	18	7	8	1
13	17	12	4	14	11	10	16	18	6	2	7	15	5	8	9	3	1
14	6	8	17	10	7	3	4	18	5	13	11	2	9	12	16	15	1
15	16	12	9	2	11	13	5	18	4	3	7	10	17	8	6	14	1
16	9	11	5	4	7	17	6	1	16	9	11	5	4	7	17	6	1
17	4	11	16	6	7	5	9	1	17	4	11	16	6	7	5	9	1
18	1	18	1	18	1	18	1	18	1	18	1	18	1	18	1	18	1

Dr. Şerif Bahtiyar

Mathematical Background

40

40

Number Theory

Discrete Logarithms

With ordinary positive real numbers, the logarithm function is the inverse of exponentiation. An analogous function exists for modular arithmetic.

Properties of ordinary logarithms

The logarithm of a number is defined to be the power to which some positive base (except 1) must be raised in order to equal the number. That is, for base x and for a value y ,

$$\begin{aligned} y &= x^{\log_x(y)} \\ \log_x(1) &= 0 \\ \log_x(x) &= 1 \\ \log_x(yz) &= \log_x(y) + \log_x(z) \\ \log_x(y^r) &= r \times \log_x(y) \end{aligned}$$

Dr. Şerif Bahtiyar

Mathematical Background

41

41

Number Theory

Discrete Logarithms

Consider a primitive root a for some prime number p . Then we know that the powers of a from 1 through $(p-1)$ produce each integer from 1 through $(p-1)$ exactly once.

We also know that any integer b satisfies

$$b \equiv r \pmod{p} \text{ for some } r, \text{ where } 0 \leq r \leq (p-1)$$

For any integer b and a primitive root a of prime number p , we can find a unique exponent i such that

$$b \equiv a^i \pmod{p} \text{ where } 0 \leq i \leq (p-1)$$

Dr. Şerif Bahtiyar

Mathematical Background

42

42

Number Theory

Discrete Logarithms

This exponent i is referred to as the **discrete logarithm** of the number b for the base $a \pmod{p}$. We denote this value as $\text{dlog}_{a,p}(b)$.¹¹

Note the following:

$$\text{dlog}_{a,p}(1) = 0 \text{ because } a^0 \bmod p = 1 \bmod p = 1 \quad (2.21)$$

$$\text{dlog}_{a,p}(a) = 1 \text{ because } a^1 \bmod p = a \quad (2.22)$$

Dr. Şerif Bahtiyar

Mathematical Background

43

Number Theory

Discrete Logarithms

Here is an example using a nonprime modulus, $n = 9$. Here $\phi(n) = 6$ and $a = 2$ is a primitive root. We compute the various powers of a and find

$$2^0 = 1 \quad 2^4 \equiv 7 \pmod{9}$$

$$2^1 = 2 \quad 2^5 \equiv 5 \pmod{9}$$

$$2^2 = 4 \quad 2^6 \equiv 1 \pmod{9}$$

$$2^3 = 8$$

This gives us the following table of the numbers with given discrete logarithms (mod 9) for the root $a = 2$:

Logarithm	0	1	2	3	4	5
Number	1	2	4	8	7	5

To make it easy to obtain the discrete logarithms of a given number, we rearrange the table:

Number	1	2	4	5	7	8
Logarithm	0	1	2	5	4	3

Dr. Şerif Bahtiyar

Mathematical Background

44

Number Theory

Discrete Logarithms

Some Properties

Any positive integer z can be expressed in the form $z = q + k\phi(n)$, with $0 \leq q < \phi(n)$. Therefore, by Euler's theorem,

$$a^z \equiv a^q \pmod{n} \quad \text{if } z \equiv q \pmod{\phi(n)}$$

Applying this to the foregoing equality, we have

$$\text{dlog}_{a,p}(xy) \equiv [\text{dlog}_{a,p}(x) + \text{dlog}_{a,p}(y)] \pmod{\phi(p)}$$

and generalizing,

$$\text{dlog}_{a,p}(y^r) \equiv [r \times \text{dlog}_{a,p}(y)] \pmod{\phi(p)}$$

Keep in mind that unique discrete logarithms mod m to some base a exist only if a is a primitive root of m .

Dr. Şerif Bahtiyar

Mathematical Background

45

Number Theory

Discrete Logarithms

Table 2.8 Tables of Discrete Logarithms, Modulo 19

(a) Discrete logarithms to the base 2, modulo 19

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\text{dlog}_{2,19}(a)$	18	1	13	2	16	14	6	3	8	17	12	15	5	7	11	4	10	9

(b) Discrete logarithms to the base 3, modulo 19

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\text{dlog}_{3,19}(a)$	18	7	1	14	4	8	6	3	2	11	12	15	17	13	5	10	16	9

(c) Discrete logarithms to the base 10, modulo 19

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\text{dlog}_{10,19}(a)$	18	17	5	16	2	4	12	15	10	1	6	3	13	11	7	14	8	9

(d) Discrete logarithms to the base 13, modulo 19

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\text{dlog}_{13,19}(a)$	18	11	17	4	14	10	12	15	16	7	6	3	1	5	13	8	2	9

(e) Discrete logarithms to the base 14, modulo 19

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\text{dlog}_{14,19}(a)$	18	13	7	8	10	2	6	3	14	5	12	15	11	1	17	16	4	9

(f) Discrete logarithms to the base 15, modulo 19

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\text{dlog}_{15,19}(a)$	18	5	11	10	8	16	12	15	4	13	6	3	7	17	1	2	14	9

Dr. Şerif Bahtiyar

Mathematical Background

46

Number Theory

Discrete Logarithms

$$y = g^x \bmod p$$

Given g , x , and p , it is a straightforward matter to calculate y . At the worst, we must perform x repeated multiplications, and **algorithms exist for achieving greater efficiency**.

However, given y , g , and p , it is, in general, **very difficult to calculate x** .

The **difficulty** seems to be on the **same order** of magnitude as that of **factoring primes required for RSA**.

Dr. Şerif Bahtiyar

Mathematical Background

47

Summary

Significant theorems used in cryptography.

Dr. Şerif Bahtiyar

Mathematical Background

48

Questions?

Dr. Şerif Bahtıyar

Mathematical Background

49